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Pretap equalizable continuous time linear equalizer

Abstract

A circuit includes first, second, third, and fourth transistors, and a capacitor. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor has a first terminal, second terminal, and a control terminal. The capacitor has a first conductor coupled to the second terminal of the first transistor, and a second conductor coupled to the second terminal of the second transistor. The third transistor has a first terminal coupled to the first terminal of the second transistor, a second terminal, and a control terminal coupled to the control terminal of the first transistor. The fourth transistor has a first terminal coupled to the first terminal of the first transistor, a second terminal coupled to the second terminal of the third transistor, and a control terminal coupled to the control terminal of the second transistor.

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Background/Summary

BACKGROUND

(1) Serial communication links experience high frequency distortion (phase and amplitude) between the transmitter and receiver over a lossy channel. This distortion is manifested at the receiver as inter-symbol interference, i.e., a smearing of the transmitted data bits/symbols. Channel equalization is used to counteract inter-symbol interference and other channel induced distortion. Channel equalization may be applied using transmission pre-emphasis that pre-distorts a transmit signal and/or as receiver equalization that applies post-compensation for the undesirable frequency effects of the channel.

SUMMARY

(2) In one example, a circuit includes first, second, third, and fourth transistors, and a capacitor. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor has a first terminal, second terminal, and a control terminal. The capacitor has a first conductor coupled to the second terminal of the first transistor, and a second conductor coupled to the second terminal of the second transistor. The third transistor has a first terminal coupled to the first terminal of the second transistor, a second terminal, and a control terminal coupled to the control terminal of the first transistor. The fourth transistor has a first terminal coupled to the first terminal of the first transistor, a second terminal coupled to the second terminal of the third transistor, and a control terminal coupled to the control terminal of the second transistor.

(3) In another example, a circuit includes first, second, third, fourth, and fifth transistors. The first

transistor has a first terminal, a second terminal, and a control terminal. The second transistor is coupled to the first transistor as a first differential pair. The second transistor has a first terminal, a second terminal, and a control terminal. The third transistor has a first terminal, a second terminal, and a control terminal. The first terminal of the third transistor is coupled to the first terminal of the second transistor. The control terminal of the third transistor is coupled to the control terminal of the first transistor. The fourth transistor is coupled to the third transistor as a second differential pair. The fourth transistor has a first terminal, a second terminal, and a control terminal. The first terminal of the fourth transistor is coupled to the first terminal of the first transistor. The control terminal of the fourth transistor is coupled to the control terminal of the second transistor. The fifth transistor is coupled to the second differential pair. The fifth transistor is configured to switchably enable pre-tap equalization.

(4) In a further example, a deserializer includes a clock data recovery circuit and a linear equalizer circuit. The linear equalizer circuit is coupled to the clock data recovery circuit. The linear equalizer circuit includes first, second, third, fourth, and fifth transistors, a capacitor, and first and second current sources. The first transistor has a first terminal, a second terminal, and a control terminal. The second transistor has a first terminal, second terminal, and a control terminal. The capacitor has a first conductor coupled to the second terminal of the first transistor, and a second conductor coupled to the second terminal of the second transistor. The first current source has a first terminal coupled to the first conductor, and a second terminal coupled to a reference voltage terminal. The second current source has a first terminal coupled to the second conductor, and a second terminal coupled to the reference voltage terminal. The third transistor has a first terminal coupled to the first terminal of the second transistor, a second terminal, and a control terminal coupled to the control terminal of the first transistor. The fourth transistor has a first terminal coupled to the first terminal of the first transistor, a second terminal coupled to the second terminal of the third transistor, and a control terminal coupled to the control terminal of the second transistor. The fifth transistor is coupled to the third transistor. The fifth transistor is configured to switchably enable pre-tap equalization.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A a block diagram of an example serial communication system.
- (2) FIG. 1B is a block diagram of an example equalizer that includes linear equalization suitable for use in the deserializer of FIG. 1.
- (3) FIG. 2 is a schematic diagram of an example continuous time linear equalizer circuit with pre-tap equalization and optional post-tap equalization suitable for use in the equalizer of FIG. 1B.
- (4) FIG. 3 is a schematic diagram of a second example continuous time linear equalizer circuit with pre-tap equalization and optional post-tap equalization suitable for use in the equalizer of FIG. 1B.
- (5) FIG. 4 is a schematic diagram of a third example continuous time linear equalizer circuit with pre-tap equalization and optional post-tap equalization suitable for use in the equalizer of FIG. 1B.
- (6) FIG. 5 is a schematic diagram of the continuous time linear equalizer circuit of FIG. 4 showing parasitic resistances associated with the tail current sources.
- (7) FIG. 6 is a schematic diagram of a fourth example continuous time linear equalizer circuit with pre-tap equalization and optional post-tap equalization suitable for use in the equalizer of FIG. 1B.

DETAILED DESCRIPTION

(8) FIG. 1A a block diagram of an example serial communication system **100**. The serial communication system **100** includes a serializer **102** and a deserializer **104** connected via a communication channel **128**. The communication channel **128** may be a wired connection, such as a coaxial cable or a twisted pair. The serializer **102** includes an encoder **106**, a parallel-to-serial

converter **108**, and a transmitter **110** for forward channel communication. Forward channel communication may include transmission of video data, audio data, control data, etc. from the serializer **102** to the deserializer **104** via the communication channel **128**. Examples of the serializer **102** may also include a receiver **112** and a decoder **114** for back channel reception. The serializer **102** receives data to be transmitted (TDATA1), and the encoder **106** encodes the data for transmission. TDATA1 may be video received from video source or any other type of data provided by a data source. The encoder **106** may, for example, encode TDATA1 for DC balance, to add clock information, and/or introduce scrambling. The parallel-to-serial converter **108** converts the encoded data received from the encoder **106** to a serial bit-stream that is driven onto the communication channel **128** by the transmitter **110**. The transmitter **110** may provide differential or single-ended drive in various examples.

(9) The deserializer **104** includes a receiver **116**, an equalizer **118**, a serial-to-parallel converter **120**, a decoder **122**, and a clock data recovery circuit **124** for forward channel reception. The deserializer **104** may also include an encoder **126** and a transmitter **130** for back channel transmission. The receiver **116** detects the signal transmitted by the serializer **102** propagated through the communication channel **128**. The equalizer **118** applies one or more equalization techniques to recover the data received from the communication channel **128**. The equalizer **118** provides equalized data to the serial-to-parallel converter **120** and the clock data recovery circuit **124**. The clock data recovery circuit **124** recovers a clock signal from the equalized data. The clock data recovery circuit **124** provides the recovered clock signal to the equalizer **118** and the serial-to-parallel converter **120**. For example, the equalizer **118** may apply the recovered clock signal to sample the equalized data. The serial-to-parallel converter **120** converts the equalized serial data to parallel symbols. The decoder **122** reverses the encoding applied in the serializer **102** to recreate the data received by the serializer **102** for transmission (to recreate TDATA1). Accordingly, the output of the decoder **122** (RDATA1) is a recreation of TDATA1. The deserializer **104** may provide RDATA1 to a video display, a video processing system, a processor, or any other circuitry configured to receive from the deserializer **104**.

(10) Back channel communication may include transmission of control data or other data from the deserializer **104** to the serializer **102** via the communication channel **128** (e.g., simultaneous with forward channel transmission via the communication channel **128**). The encoder **126** encodes data (e.g., TDATA2) received for transmission. TDATA2 may be a serial data stream received from a processor or other device. The transmitter **130** drives the encoded data received from the encoder **126** onto the communication channel **128**.

(11) In the serializer **102**, the receiver **112** receives the signal present on the communication channel **128** and separates the data transmitted by the transmitter **130** from the data transmitted by the transmitter **110**. The receiver **112** provides the received back channel data to the decoder **114**, and the decoder **114** reverses the encoding applied in the deserializer **104** to recreate the data received by the deserializer **104** for transmission (to recreate TDATA2). Accordingly, the output of the decoder **114** (RDATA2) is a recreation of TDATA2. RDATA2 may be provided to processor or other device.

(12) FIG. 1B is a block diagram of an example equalizer **150**. The equalizer **150** may be an example of the equalizer **118**. The equalizer **150** includes a continuous time linear equalizer circuit **152**. Some examples of the equalizer **150** may also include a decision feedback equalizer **156**, a feed forward equalizer, or other equalizer circuit. A clock data recovery circuit **154** and a processor **158** may be coupled to the equalizer **150**. The continuous time linear equalizer circuit **152** is a linear equalizer that boosts both signal and noise. The decision feedback equalizer **156** is coupled to an output of the continuous time linear equalizer circuit **152**. The decision feedback equalizer **156** is a non-linear equalizer that is immune to noise, but may present feedback loop timing challenges. The clock data recovery circuit **154** extracts a clock signal from the output of the continuous time linear equalizer circuit **152** for use by the decision feedback equalizer **156** and/or

other circuitry (e.g., the serial-to-parallel converter **120**).

(13) The decision feedback equalizer **156** may provide post-tap equalization (correction of post-cursor inter-symbol interference). As data rate and modulation complexity increase, e.g., with implementation of pulse-amplitude modulation 4-level (PAM4), pre-tap equalization (correction of pre-cursor inter-symbol interference) becomes desirable. In some systems, a feed-forward equalizer in discrete time may be used to provide pre-tap equalization. Discrete time feed-forward equalizers can be costly in terms of both power and circuit area. The equalizer **150** includes a post-tap equalization circuit **160**, a pre-tap equalization circuit **162**, and a pre-tap enable circuit **164**. The post-tap equalization circuit **160** provides post-tap equalization. The pre-tap equalization circuit **162** provides pre-tap equalization. The pre-tap enable circuit **164** controls the post-tap equalization circuit **160** and/or the pre-tap equalization circuit **162** to select pre or post-tap equalization based on a control signal received from the processor **158** or other control circuit. In the equalizer **150**, examples of the continuous time linear equalizer circuit **152** may provide pre-tap equalization with reduced power consumption and circuit area relative to other pre-tap equalization circuits.

(14) The unit interval sampled pulse response of a continuous time linear equalizer output has a time domain response given by the finite impulse response:

$$H_{\text{post}}(z) = a - bZ^{-1} \quad (1)$$

where a is the coefficient of the main tap and b is the coefficient of the post tap.

(15) The continuous time equivalent of $H(Z)$, ($H(s)$), can be found using inverse bilinear transformation:

$$(16) \quad Z = e^{sT} = \frac{1 + sT/2}{1 - sT/2} \quad (2)$$

where $T = 1/f_s$ (baud rate).

$$(17) \quad H_{\text{post}}(s) = (a - b) \left(\frac{1 + \frac{s(a+b)T}{2}}{1 - \frac{s(a-b)T}{2}} \right) \quad (3)$$

(18) Equation (3) shows that the left half plane zero generates a post-tap. The ratio of pole to zero

$$(19) \quad \left(\frac{a+b}{a-b} \right)$$

determines the boost in frequency and the values of the main tap coefficient a and the post-tap coefficient b .

(20) To realize a pre-tap:

$$H_{\text{pre}}(z) = -b + aZ^{-1} \quad (4)$$

where the main tap has a coefficient a , and the pre-tap has a coefficient b .

(21) Using bilinear transformation:

$$(22) \quad H_{\text{pre}}(s) = (a - b) \left(\frac{1 - \frac{s(a+b)T}{2}}{1 + \frac{s(a-b)T}{2}} \right) \quad (5)$$

(23) Comparing equations (3) and (5), the difference between H_{pre} and H_{post} is the presence of a right half plane zero in H_{pre} .

(24) The post/pre-tap pulse response for a continuous time linear equalizer may be defined as:

$$H_{\text{ctle}}.\text{sub.pulse}(t) = (U(t) - U(t-T)) \text{conv } H_{\text{ctle}}(t) \quad (6)$$

where $U(t)$ is unit step, T is pulse width corresponding to baud rate, and $H_{\text{ctle}}(t)$ is the impulse response of the continuous time linear equalizer.

(25) In the Laplace domain:

$$(26) \quad H_{\text{ctle_post}}_{\text{pulse}(s)} = \left(\frac{1}{s} (1 - e^{-sT}) \right) X H_{\text{post}}(s) \quad (7) \quad \text{where } H_{\text{post}}(s) = \left(\frac{1 + \frac{s}{P}}{1 + \frac{s}{Q}} \right). \quad (8)$$

$$H_{\text{ctle_post}}_{\text{pulse}(t)} = [(e^{-Pt}) \left(\frac{P}{Z} - 1 \right) + 1] u(t) - [(e^{-P(t-T)}) \left(\frac{P}{Z} - 1 \right) + 1] u(t-T) \quad (9)$$

$$H_{\text{ctle_pre}}_{\text{pulse}(s)} = \left(\frac{1}{s} (1 - e^{-sT}) \right) X H_{\text{pre}}(s) \quad (10) \quad \text{where } H_{\text{pre}}(s) = \left(\frac{1 + \frac{s}{Q}}{1 + \frac{s}{P}} \right).$$

$$H_{\text{ctle_pre}}_{\text{pulse}(t)} = [1 - (e^{-Pt}) \left(\frac{P}{Z} + 1 \right)] u(t) - [1 - (e^{-P(t-T)}) \left(\frac{P}{Z} + 1 \right)] u(t-T) \quad (11)$$

(27) A pre-tap continuous time linear equalizer may be derived from the post-tap continuous time linear equalizer of equation (8).

$$(28) \quad H_{pre}(s) = 2 - H_{post}(s) = 2 - \left(\frac{1 + \frac{s}{Z}}{1 + \frac{s}{P}} \right) \quad (12)$$

(29) With zero $Z=P/K$, (boost of post tap continuous time linear equalizer= K)

$$(30) \quad H_{pre}(s) = \left(\frac{1 - \frac{s}{P}(k-2)}{1 + \frac{s}{P}} \right) \quad (13)$$

(31) In equation (13), the right hand plane zero provides a pre-tap boost of $K-2$. $Z=P/(k+2)$ is needed to provide an equivalent post-tap continuous time linear equalizer boost.

(32) FIG. 2 is a schematic diagram of an example continuous time linear equalizer circuit **200** with pre-tap equalization suitable for use in the equalizer **150**. The continuous time linear equalizer circuit **200** is an example of the continuous time linear equalizer circuit **152**. The continuous time linear equalizer circuit **200** includes transistors **202** and **204** connected as a first differential pair, and transistors **206** and **208** connected as a second differential pair. A first current terminal (e.g., drain) of the transistor **202** is coupled to a first output terminal (VOUTP) and to a first current terminal (e.g., drain) of the transistor **208**. A first current terminal (e.g., drain) of the transistor **204** is coupled to a second output terminal (VOUTM) and to a first current terminal (e.g., drain) of the transistor **206**. A control terminal (e.g., gate) of the transistor **202** is coupled to a first input terminal (VINP) and to a control terminal (e.g., gate) of the transistor **206**. A control terminal (e.g., gate) of the transistor **204** is coupled to a second input terminal (VINM) and to a control terminal (e.g., gate) of the transistor **208**. A second current terminal (e.g., source) of the transistor **206** is coupled to a second control terminal (e.g., source) of the transistor **208**.

(33) The continuous time linear equalizer circuit **200** also includes a transistor **210**, current sources **212**, **216**, and **218**, a resistor **220**, and a capacitor **214**. A first current terminal (e.g., drain) of the transistor **210** is coupled to the second current terminals of the transistor **206** and the transistor **208**. A control terminal (e.g., gate) of the transistor **210** is coupled to a pre-tap equalization enable terminal (EN_PRE). A second current terminal (e.g., source) of the transistor **210** is coupled to a first terminal of the current source **212**. A second terminal of the current source **212** is coupled to a reference voltage terminal (e.g., ground). The transistor **210** may be turned on to switchably enable pre-tap equalization (activate tail current flow through the transistors **206** and **208**) in the continuous time linear equalizer circuit **200**, and turned off to disable pre-tap equalization. The transistors **202**, **204**, **206**, **208**, and **210** may be n-type field effect transistors.

(34) A first terminal of the resistor **220** is coupled to a second current terminal (e.g., source) of the transistor **202**. A second terminal of the resistor **220** is coupled to a second current terminal (e.g., source) of the transistor **204**. The capacitor **214** is coupled to the transistor **202** and the transistor **204** in parallel with the resistor **220**. A first conductor (e.g., top plate) of the capacitor **214** is coupled to the first terminal of the resistor **220**, and a second conductor (e.g., bottom plate) of the capacitor **214** is coupled to the second terminal of the resistor **220**.

(35) The current source **216** is coupled between the second current terminal of the transistor **202** and the reference voltage terminal. The current source **218** is coupled between the second current terminal of the transistor **204** and the reference voltage terminal. A first terminal of the current source **216** is coupled to the second current terminal of the transistor **202**, and a second terminal of the current source **216** is coupled to the reference voltage terminal. A first terminal of the current source **218** is coupled to the second current terminal of the transistor **204**, and a second terminal of the current source **218** is coupled to the reference voltage terminal.

(36) In the continuous time linear equalizer circuit **200**, the transistor **210** is turned off (EN_PRE=0) to activate post-tap equalization and the transistor **210** is turned on (EN_PRE=1) to activate pre-tap equalization. If the transistor **210** is turned off,

$$(37) \quad H_{tle_post}(s) = H_{tle}(s) = \left(\frac{gm}{1 + gm * Rs/2} \right) \left(\frac{1 + S(Rs * Cs)}{1 + \frac{S * Rs * Cs}{(1 + gm * \frac{Rs}{2})}} \right) * Z_{out} \quad (14)$$

(38) With

$$(39) \quad dcgain = \frac{gm}{1 + gm * Rs/2}, p = \frac{(1 + gm * \frac{Rs}{2})}{Rs * Cs}, \text{ and } z = \frac{(1)}{Rs * Cs} = \frac{P}{1 + gm * \frac{Rs}{2}},$$

the post boost is

$$(40) \quad 0 \leq K = 1 + gm * \frac{Rs}{2} \quad (15)$$

(41) If the transistor **210** is turned on, the transistor **206** and the transistor **208** turn on with:

$$(42) \quad GM_PRE = 2 * \left(\frac{gm}{1 + gm * Rs/2} \right) \quad (16)$$

$$H_{ctle_pre}(s) = \left(\frac{gm}{1 + gm * Rs/2} \right) \left(\frac{1 - \frac{S * Rs * Cs}{(1 + gm * \frac{Rs}{2})} * (gm * \frac{Rs}{2} - 1)}{1 + \frac{S * Rs * Cs}{(1 + gm * \frac{Rs}{2})}} \right) * Z_{out} \quad (17)$$

(43) Pre-tap continuous time linear equalizer boost is:

$$(44) \quad (gm * \frac{Rs}{2} - 1) = k - 2 \quad (18)$$

(45) In some examples, the transistor **206** may be 2/K times the size of the transistor **202**.

(46) FIG. 3 is a schematic diagram of a second example continuous time linear equalizer circuit **300** with pre-tap equalization suitable for use in the equalizer **150**. The continuous time linear equalizer circuit **300** is an example of the continuous time linear equalizer circuit **152**. The continuous time linear equalizer circuit **300** may provide greater pre-tap equalization boost than the continuous time linear equalizer circuit **200**, and may consume less circuit area. The continuous time linear equalizer circuit **300** includes transistors **202**, **204**, **206**, and **208**, capacitor **214**, and current sources **212**, **216**, and **218** similar to the continuous time linear equalizer circuit **200**. The continuous time linear equalizer circuit **300** also includes transistors **302**, **304**, **306**, and **308** that select the connection of the transistor **206** and the transistor **208** to the transistor **202** and the transistor **204**. Pre-tap equalization is selected by turning on the transistor **302** and the transistor **304** (and turning off the transistor **306** and the transistor **308**). Post-tap equalization is selected by turning on the transistor **306** and the transistor **308** (and turning off the transistor **302** and the transistor **304**).

(47) The transistors **302**, **304**, **306**, and **308** may be n-type FETs. The transistor **302** includes a first current terminal (e.g., drain) coupled to the first current terminal of the transistor **204**, and a second current terminal (e.g., source) coupled to the first current terminal of the transistor **206**. A control terminal (e.g., gate) of the transistor **302** is coupled to a pre-tap equalization enable terminal (EN_PRE). The transistor **304** includes a first current terminal (e.g., drain) coupled to the first current terminal of the transistor **202**, and a second current terminal (e.g., source) coupled to the first current terminal of the transistor **208**. A control terminal (e.g., gate) of the transistor **304** is coupled to the control terminal of the transistor **302**.

(48) The transistor **306** includes a first current terminal (e.g., drain) coupled to the first current terminal of the transistor **202**, and a second current terminal (e.g., source) coupled to the first current terminal of the transistor **206**. A control terminal (e.g., gate) of the transistor **306** is coupled to a post-tap equalization enable terminal (EN_PRE). The transistor **308** includes a first current terminal (e.g., drain) coupled to the first current terminal of the transistor **204**, and a second current terminal (e.g., source) coupled to the first current terminal of the transistor **208**. A control terminal (e.g., gate) of the transistor **308** is coupled to the control terminal of the transistor **306**.

(49) The transfer function of the continuous time linear equalizer circuit **300** with pre-tap equalization may be:

$$(50) \quad H_{ctle_pre}(s) = (gm_pre) \left(\frac{1 - SCs * \frac{gm - gm_pre}{gm * gm_pre}}{1 + \frac{S * Cs}{gm}} \right) * Z_{out} \quad (19)$$

(51) Pre-tap equalization boost and transconductance may be defined as:

$$(52) \quad \text{Boost} = \text{Zero} / \text{pole} = \left(\frac{gm}{gm_pre} - 1 \right) \quad (20) \quad gm_pre = \frac{gm}{1 + gm * Rs} = gm * Rs = k - 1 \quad (21)$$

(53) As per equation (21), the continuous time linear equalizer circuit **300** may provide a pre-tap equalization boost of k-1, rather than k-2 as in the continuous time linear equalizer circuit **200**.

(54) FIG. 4 is a schematic diagram of a third example continuous time linear equalizer circuit **400** with pre-tap equalization suitable for use in the equalizer **150**. The continuous time linear equalizer circuit **400** is an example of the continuous time linear equalizer circuit **152**. The continuous time

linear equalizer circuit **400** may provide greater pre-tap equalization boost than the continuous time linear equalizer circuit **200**, and may consume less circuit area. The continuous time linear equalizer circuit **400** includes transistors **202**, **204**, **206**, **208**, and **210**, capacitor **214**, and current sources **212**, **216**, and **218** similar to the continuous time linear equalizer circuit **200**. The continuous time linear equalizer circuit **400** also includes resistors **402** and **404**, and transistor **406**. The transistor **406** connects the resistors **402** and **404** in parallel with the capacitor **214**. A first terminal of the resistor **402** is coupled to the second terminal of the transistor **202**, and a second terminal of the **402** is coupled to a first terminal (e.g., drain) of the transistor **406**. A first terminal of the resistor **404** is coupled to the second terminal of the transistor **204**, and a second terminal of the resistor **404** is coupled to a second terminal (e.g., source) of the transistor **406**. A control terminal (e.g., gate) of the transistor **406** is coupled to pre-tap equalization disable terminal (EN_PREZ). Pre-tap equalization is selected by turning on the transistor **210** and turning off the transistor **406**. Post-tap equalization is selected by turning on the transistor **406** and turning off the transistor **210**.

(55) With pre-tap equalization enabled, the performance of the continuous time linear equalizer circuit **400** is similar to that of the continuous time linear equalizer circuit **300**. With pre-tap equalization disabled (post-tap equalization enabled), the performance of the continuous time linear equalizer circuit **400** is similar to the continuous time linear equalizer circuit **200** in post-top equalization mode. In the continuous time linear equalizer circuit **300**, voltage headroom is consumed by the transistors **302**, **304**, **306**, and **308**. This causes linearity degradation in the output of the continuous time linear equalizer circuit **300**. Accordingly, the continuous time linear equalizer circuit **400** may provide improved linearity relative to the continuous time linear equalizer **300**.

(56) FIG. 5 is a schematic diagram of the continuous time linear equalizer circuit **400** showing parasitic resistors **502** and **504** in parallel with the current source **216** and the current source **218**. With pre-tap equalization enabled, the resistors **502** and **504** (resistances R_d) across the current sources **216** and **218** draw a current in opposite phase to

$$(57) \text{ gm_pre } \cdot \text{Math. gmd} = \frac{\text{gm}}{1 + \text{gm}R_d}$$

that may degrade equalizer performance.

(58) At low frequencies, the net current is $G_{m_dc} = (\text{gm_pre} - \text{gmd})$. This DC gain is compromised due to the current flowing in the opposite phase. To maintain reasonable DC gain, $G_{m_pre_new} = \text{gm_pre} + \text{gmd}$. This restores $G_{m_dc} = \text{gm_pre}$.

(59) At high frequencies, the capacitor **214** bypasses the resistors **502** and **504**.

$$G_{m_high_freq} = \text{gm} - \text{gm_pre_new} \quad (22)$$

$$G_{m_high_freq} = \text{gm} - \text{gm_pre} - \text{gmd} \quad (23)$$

$$(60) \text{ Boost} = \frac{G_{m_high_freq}}{G_{m_DC}} = \frac{\text{gm} \cdot \text{Math. gm_pre} - \text{gmd}}{\text{gm_pre}} = K - 1 - \frac{\text{gmd}}{\text{gm_pre}} \quad (24)$$

$$\text{where } \text{gmd} = \frac{\text{gm}}{1 + \text{gm}R_d}, \text{ gm_pre} = \frac{\text{gm}}{1 + \text{gm} \cdot \frac{R_s}{2}}, \text{ Boost} = K - 1 - \frac{1 + \text{gm} \cdot \frac{R_s}{2}}{1 + \text{gm}R_d} = K - 1 - R_s / 2R_d \quad (25)$$

(61) FIG. 6 is a schematic diagram of a fourth example continuous time linear equalizer circuit **600** with pre-tap equalization suitable for use in the equalizer **150**. The continuous time linear equalizer circuit **600** is similar to the continuous time linear equalizer circuit **400**. The continuous time linear equalizer circuit **600** includes transistors **202**, **204**, **206**, **208**, **210**, and **406**, capacitor **214**, resistors **402** and **404**, and current sources **212**, **216**, and **218** similar to the continuous time linear equalizer circuit **400**. The continuous time linear equalizer circuit **600** also includes a current source **602**, and transistors **604** and **606**. The transistors **604** and **606** may be p-type FETs. An input terminal of the current source **602** is coupled to a power supply terminal (VDD). A first current terminal (e.g., source) of the transistor **604** is coupled to an output terminal of the current source **602**, and a second current terminal (e.g., drain) of the transistor **604** is coupled to second current terminal of the transistor **204**. A control terminal (e.g., gate) of the transistor **604** is coupled to the second current terminal of the transistor **202**. A first current terminal (e.g., source) of the transistor **606** is

coupled to the output terminal of the current source **602**, and a second current terminal (e.g., drain) of the transistor **606** is coupled to second current terminal of the transistor **202**. A control terminal (e.g., gate) of the transistor **606** is coupled to the second current terminal of the transistor **204**.

(62) The transistors **604** and **606** operate as gds cancellation transistors. The transistors **604** and **606** increase the effective resistance of the resistor **502** and the resistor **504** as:

$$R_{\text{deff}} = R_d / (1 - R_d * G_{\text{mcan}}) \quad (26)$$

where G_{mcan} is the gm of the transistors **604** and **606**.

(63) The transistors **604** and **606** reduce the DC output of phase current with respect to $g_{\text{m_pre}}$ to produce a pre-tap boost that is very close to $K-1$ in the continuous time linear equalizer circuit **600**.

(64) While the continuous linear equalizer circuits **200**, **300**, **400**, and **600** have been described herein as including n-type FETs, some implementations of the continuous linear equalizer circuits **200**, **300**, **400**, or **600** may be implemented using p-type FETs and/or complementary metal oxide semiconductor (CMOS) input structures.

(65) In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

(66) Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

(67) A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

(68) As used herein, the terms “terminal,” “node,” “interconnection,” “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

(69) A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

(70) While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor (“FET”) (such as an n-channel FET (NFET) or a p-channel FET (PFET)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors, or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate

(GaAs).

(71) References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

(72) References herein to a FET being “ON” or “enabled” means that the conduction channel of the FET is present and drain current may flow through the FET. References herein to a FET being “OFF” or “disabled” means that the conduction channel is not present so drain current does not flow through the FET. An “OFF” FET, however, may have current flowing through the transistor's body-diode.

(73) Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

(74) While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

(75) Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

(76) Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A circuit comprising: a first transistor having a first terminal, a second terminal, and a control terminal; a second transistor having a first terminal, second terminal, and a control terminal; a capacitor having a first conductor coupled to the second terminal of the first transistor, and a second conductor coupled to the second terminal of the second transistor; a third transistor having: a first terminal coupled to the first terminal of the second transistor; a second terminal; and a control terminal coupled to the control terminal of the first transistor; and a fourth transistor having: a first terminal coupled to the first terminal of the first transistor; a second terminal coupled to the second terminal of the third transistor; and a control terminal coupled to the control terminal of the second transistor.
2. The circuit of claim 1, further comprising: a fifth transistor having: a first terminal coupled to the second terminal of the fourth transistor; a second terminal; and a control terminal coupled to a pre-

- tap equalization enable terminal; and a current source having: a first terminal coupled to the second terminal of the fifth transistor; and a second terminal coupled to a reference voltage terminal.
3. The circuit of claim 1, further comprising: a resistor having: a first terminal coupled to the second terminal of the first transistor; and a second terminal coupled to the second terminal of the second transistor; a first current source having: a first terminal coupled to the second terminal of the first transistor; and a second terminal coupled to a reference voltage terminal; and a second current source having: a first terminal coupled to the second terminal of the second transistor; and a second terminal coupled to a reference voltage terminal.
4. The circuit of claim 1, further comprising: a fifth transistor having a first terminal, a second terminal, and a control terminal the control terminal coupled to a pre-tap equalization disable terminal; a first resistor having: a first terminal coupled to the second terminal of the first transistor; and a second terminal coupled to the first terminal of the fifth transistor; a second resistor having: a first terminal; and coupled to the second terminal of the fifth transistor; and a second terminal coupled to the second terminal of the second transistor; a first current source having: a first terminal coupled to the second terminal of the first transistor; and a second terminal coupled to a reference voltage terminal; and a second current source having: a first terminal coupled to the second terminal of the second transistor; and a second terminal coupled to the reference voltage terminal.
5. The circuit of claim 3, further comprising: a fifth transistor having: a first terminal; a second terminal coupled to first terminal of the second current source; and a control terminal coupled to the second terminal of the first transistor; a sixth transistor having: a first terminal coupled to the first terminal of the fifth transistor; a second terminal coupled to the first terminal of the first current source; and a control terminal coupled to the second terminal of the second transistor; and a third current source having: a first terminal coupled to a power supply terminal; and a second terminal coupled to the first terminal of the fifth transistor.
6. The circuit of claim 1, further comprising: a fifth transistor having: a first terminal coupled to the first terminal of the second transistor; a second terminal coupled to the first terminal of the third transistor; and a control terminal; a sixth transistor having: a first terminal coupled to the first terminal of the first transistor; a second terminal coupled to the first terminal of the fourth transistor; and a control terminal coupled to the control terminal of the fifth transistor; a seventh transistor having: a first terminal coupled to the first terminal of the sixth transistor; a second terminal coupled to the second terminal of the fifth transistor; and a control terminal; an eighth transistor having: a first terminal coupled to the first terminal of the fifth transistor; a second terminal coupled to the second terminal of the sixth transistor; and a control terminal coupled to the control terminal of the seventh transistor.
7. A circuit comprising: a first transistor having a first terminal, a second terminal, and a control terminal; a second transistor coupled to the first transistor as a first differential pair, the second transistor having a first terminal, a second terminal, and a control terminal; a third transistor having: a first terminal coupled to the first terminal of the second transistor; a second terminal; and a control terminal coupled to the control terminal of the first transistor; a fourth transistor coupled to the third transistor as a second differential pair, the fourth transistor having: a first terminal coupled to the first terminal of the first transistor; a second terminal; and a control terminal coupled to the control terminal of the second transistor; and a fifth transistor coupled to the second differential pair, the fifth transistor configured to switchably enable pre-tap equalization.
8. The circuit of claim 7, further comprising: a current source having: a first terminal; and a second terminal coupled to a reference voltage terminal; wherein: the fifth transistor has: a first terminal coupled to the second terminal of the third transistor and the second terminal of the fourth transistor; a second terminal coupled to the first terminal of the current source; and a control terminal coupled to a pre-tap equalization enable terminal.
9. The circuit of claim 7, further comprising: a sixth transistor having: a first terminal coupled to

the first terminal of the first transistor; a second terminal coupled to the first terminal of the fourth transistor; and a control terminal; a seventh transistor having: a first terminal coupled to the first terminal of the sixth transistor; a second terminal coupled to the first terminal of the third transistor; and a control terminal; an eighth transistor having: a first terminal coupled to the first terminal of the second transistor; a second terminal coupled to the second terminal of the sixth transistor; and a control terminal coupled to the control terminal of the seventh transistor; wherein the fifth transistor has: a first terminal coupled to the first terminal of the second transistor; a second terminal coupled to the first terminal of the third transistor; and a control terminal coupled to the control terminal of the sixth transistor.

10. The circuit of claim 7, further comprising a capacitor having: a first conductor coupled to the second terminal of the first transistor; and a second conductor coupled to the second terminal of the second transistor.

11. The circuit of claim 10, further comprising: a first current source having: a first terminal coupled to the first conductor; a second terminal coupled to a reference voltage terminal; and a second current source having: a first terminal coupled to the second conductor; and a second terminal coupled to the reference voltage terminal.

12. The circuit of claim 7, further comprising a resistor having: a first terminal coupled to the second terminal of the first transistor; and a second terminal coupled to the second terminal of the second transistor.

13. The circuit of claim 7, further comprising: a first resistor having: a first terminal coupled to the second terminal of the first transistor; and a second terminal; a second resistor having: a first terminal coupled to the second terminal of the second transistor; and a second terminal; a sixth transistor having: a first terminal coupled to the second terminal of the first resistor; a second terminal coupled to the second terminal of the second resistor; and a control terminal coupled to a pre-tap equalization disable terminal.

14. The circuit of claim 13, further comprising: a current source having: a first terminal coupled to a power supply terminal; and a second terminal; a seventh transistor having: a first terminal coupled to the second terminal of the current source; a second terminal coupled to the second terminal of the second transistor; and a control terminal coupled to the second terminal of the first transistor; and a eighth transistor having: a first terminal coupled to the second terminal of the current source; a second terminal coupled to the second terminal of the first transistor; and a control terminal coupled to the second terminal of the second transistor.

15. A deserializer comprising: a clock data recovery circuit; a linear equalizer circuit coupled to the clock data recovery circuit, the linear equalizer circuit including: a first transistor having a first terminal, a second terminal, and a control terminal; a second transistor having a first terminal, second terminal, and a control terminal; a capacitor having: a first conductor coupled to the second terminal of the first transistor; and a second conductor coupled to the second terminal of the second transistor; a first current source having: a first terminal coupled to the first conductor; and a second terminal coupled to a reference voltage terminal; a second current source having: a first terminal coupled to the second conductor; and a second terminal coupled to the reference voltage terminal; a third transistor having: a first terminal coupled to the first terminal of the second transistor; a second terminal; and a control terminal coupled to the control terminal of the first transistor; and a fourth transistor having: a first terminal coupled to the first terminal of the first transistor; a second terminal coupled to the second terminal of the third transistor; and a control terminal coupled to the control terminal of the second transistor; and a fifth transistor coupled to the third transistor, the fifth transistor configured to switchably enable pre-tap equalization.

16. The deserializer of claim 15, wherein: the linear equalizer circuit includes: a current source having: a first terminal; and a second terminal coupled to a reference voltage terminal; and the fifth transistor has: a first terminal coupled to the second terminal of the fourth transistor; a second terminal coupled to the first terminal of the current source; and a control terminal coupled to a pre-

tap equalization enable terminal.

17. The deserializer of claim 15, wherein: the fifth transistor has: a first terminal coupled to the first terminal of the second transistor; a second terminal coupled to the first terminal of the third transistor; and a control terminal; and the linear equalizer circuit includes: a sixth transistor having: a first terminal coupled to the first terminal of the first transistor; a second terminal coupled to the first terminal of the fourth transistor; and a control terminal coupled to the control terminal of the fifth transistor; a seventh transistor having: a first terminal coupled to the first terminal of the sixth transistor; a second terminal coupled to the second terminal of the fifth transistor; and a control terminal; an eighth transistor having: a first terminal coupled to the first terminal of the fifth transistor; a second terminal coupled to the second terminal of the sixth transistor; and a control terminal coupled to the control terminal of the seventh transistor.

18. The deserializer of claim 15, wherein the linear equalizer circuit includes a resistor having: a first terminal coupled to the first conductor of the capacitor; and a second terminal coupled to the second conductor of the capacitor.

19. The deserializer of claim 15, wherein the linear equalizer circuit includes: a first resistor having: a first terminal coupled to the second terminal of the first transistor; and a second terminal; a second resistor having: a first terminal coupled to the second terminal of the second transistor; and a second terminal; a sixth transistor having: a first terminal coupled to the second terminal of the first resistor; a second terminal coupled to the second terminal of the second resistor; and a control terminal coupled to a pre-tap equalization disable terminal.

20. The deserializer of claim 15, wherein the linear equalizer circuit includes: a third current source having: a first terminal coupled to a power supply terminal; and a second terminal; a sixth transistor having: a first terminal coupled to the second terminal of the third current source; a second terminal coupled to the first terminal of the second current source; and a control terminal coupled to the first terminal of the first current source; and a seventh transistor having: a first terminal coupled to the second terminal of the third current source; a second terminal coupled to the first terminal of the first current source; and a control terminal coupled to the first terminal of the second current source.
